Industrial Machine Reliability & MRO Spend Analytics

Executive Assessment of Maintenance Spend, Asset Reliability, & Telemetry Signals

Prepared For: Plant Ops & Maintenance Leadership

Analysis Period: 03 Jan 2022 – 01 Jan 2025

Dataset Scope: 219,000 Records | 10 Assets | 3 Plants

1. EXECUTIVE SUMMARY & KEY VITAL SIGNS

TOTAL MRO SPEND

₹133.4 Lakh

Driven by 20,953 issue transactions across 3 years (₹1,33,40,945).

OVERALL BREAKDOWN RATE

9.88%

21,636 broken-down machine-days out of 219,000 total days examined.

TOP CATEGORY SPEND

₹32.6 Lakh

Bearings represent 24.4% of the entire MRO budget.

Over the 3-year analysis period, MRO spend and breakdown rates remained broadly stable month-on-month across the fleet. However, internal reliability is highly uneven. Critical Tier A assets and specific equipment classes (Hydraulic Presses) suffer from unacceptable failure rates. This drives substantial reactive spend and highlights that current Preventive Maintenance (PM) practices are failing to offset operational risks.

2. FINANCIAL & SPEND DIAGNOSTICS

Plant-Level Performance Comparison

Plant Location	Assets Tracked	Total Spend (₹)	Breakdown Rate (%)	Diagnostic Status
DHR-03 (Dharuhera)	3	~₹40 Lakh	**11.00%**	Red Zone: High Failure
PUN-01 (Pune)	4	~₹53.4 Lakh	9.50%	Highest Total Spend
CHN-02 (Chennai)	3	~₹40 Lakh	8.80%	Baseline Operational

Key Insight: Dharuhera Reliability Anomaly

Dharuhera (DHR-03) is the least reliable plant on a per-asset basis with an 11.00% breakdown rate, despite lower total spend than Pune. The root issue at DHR-03 is asset operational reliability rather than under-investment in spare parts.

MRO Spend Concentration by Part Family

46.8% of total MRO capital is consumed by just two part families (Bearings and Seals & Gaskets), creating a clear target for supplier renegotiation and Reliability-Centered Maintenance (RCM) reviews:

- **Bearings:** ₹32.6 Lakh (24.4%)
- **Seals & Gaskets:** ₹29.9 Lakh (22.4%)
- **Drive Belts:** ₹21.5 Lakh (16.1%)
- **Filters:** ₹20.3 Lakh (15.2%)
- **Lubrication:** ₹9.9 Lakh (7.4%)
- **Others (Electrical, Sensors, Fasteners, Couplings):** ₹19.2 Lakh (14.4%)

Top Single Cost Driver: SKU MRO-20031

Hydraulic Seal Kit 50mm Bore (SKU MRO-20031) accounts for **₹28.5 Lakh (21.3%)** of the total MRO budget alone, with 890 units issued at an average unit cost of ~₹3,200. This single SKU represents a major risk concentration requiring an immediate commercial and quality audit.

3. RELIABILITY & CRITICALITY DIAGNOSTICS

Breakdown Rate & Spend by Machine Type

Machine Type	Breakdown Rate	Total Spend (₹)
Hydraulic Press	11.32%	₹26,32,803
Screw Compressor	10.64%	₹27,14,335
EOT Crane	9.81%	₹26,42,723
CNC Lathe	9.09%	₹26,05,562
Belt Conveyor	8.53%	₹27,45,522

The Criticality Anomaly (Tier A vs C)

Criticality Tier	Breakdown Rate	Total Spend (₹)
Tier A (Critical)	12.79%	₹68.8 Lakh
Tier B (Important)	9.51%	₹49.3 Lakh
Tier C (Routine)	7.04%	₹15.2 Lakh

Criticality Paradox: Tier A critical assets experience an **82% higher failure rate** (12.79%) than Tier C routine assets (7.04%). Experiencing a >12% failure rate (>1-in-8 days) on mission-critical equipment demonstrates that conventional preventive maintenance intensity is failing to mitigate operational risk.

High-Risk Asset Target Profiles

- **AST-2017 (Hydraulic Press | PUN-01):** 12.09% Failure Rate
- **AST-5003 (EOT Crane | DHR-03):** 11.71% Failure Rate
- **AST-4055 (Screw Compressor | DHR-03):** 11.68% Failure Rate

4. PREDICTIVE CONDITION SIGNALS & SENSOR ANALYSIS

Analysis of real-time machine condition sensors during normal operation vs. breakdown events reveals critical physical leading indicators:

Sensor Parameter	Normal Baseline	Breakdown Event	Delta / Shift	Predictive Significance
Power Consumption (kW)	26.13 kW	29.40 kW	**+12.5%**	**High (Primary Signal)**
Load Percentage (%)	67.92%	72.18%	**+6.3%**	**High (Primary Signal, Load > 75%)**
Vibration - Horizontal	3.85 mm/s	4.02 mm/s	**+4.4%**	Moderate Warning
Vibration - Vertical	2.89 mm/s	3.02 mm/s	**+4.5%**	Moderate Warning
Shaft Speed (RPM)	1550.7 RPM	1580.4 RPM	+1.9%	Minor Correlation
Oil Pressure	5.00 bar	5.08 bar	+1.6%	Minor Correlation
Motor Temperature	71.06 °C	71.33 °C	+0.3%	Negligible / Uninformative
Bearing Temperature	65.08 °C	65.25 °C	+0.3%	Negligible / Uninformative

Critical Technical Finding: Temperature Blindness

> Electrical and mechanical loading stress (***Power*** and ***Load %***)—**not thermal drift**—are the true early-warning signals for failure in this fleet. Temperature sensors exhibit almost zero delta prior to breakdown (+0.3%) and are functionally blind to imminent operational failures.

Multi-Variate Correlation Mapping: Individual sensors rarely cross static SCADA threshold limits prior to failure. Instead, breakdowns occur during combined multi-variate stress events, specifically where sustained ***Load % > 75\%*** intersects with spikes in ***Power (kW)*** and elevated vibration.

5. STRATEGIC RECOMMENDATIONS & ACTION BLUEPRINT

Phase	Strategic Focus	Specific Core Action & Deliverable
Phase 1: Operational	Condition-Based Maintenance (CBM) Pilot	Deploy a multi-variate early-warning algorithm (combining Load, Power, and Vibration) on high-risk assets: **AST-2017, AST-5003, and AST-4055**. Replace single-sensor threshold alarms.
Phase 2: Tactical	Dharuhera Audit & Tier A Overhaul	Conduct a root-cause maintenance audit at **DHR-03**. Increase PM intensity and quality controls for Tier A assets to reduce failure rates from 12.79% toward the 7.04% Tier C baseline.
Phase 3: Financial	Targeted Commercial Procurement	Renegotiate vendor contracts and bulk pricing specifically for **Bearings, Seals & Gaskets**, and SKU **MRO-20031 (Hydraulic Seal Kit)** to curb the 46.8% spend bleed. Track monthly breakdown-spend reduction as ROI.

6. DATA ARCHITECTURE & METHODOLOGY

Dataset & Validation: Analysis covers 219,000 continuous machine-day rows across 1,094 days (Jan 2022 – Jan 2025). Missing values were restricted strictly to work-order fields (structurally expected on non-breakdown days). Sensor ranges were validated within physical industrial parameters (***Temp: 43.5 - 88.4^\circ ext{C}***, ***Vibration: 0.4 - 13.3 ext{ mm/s}***) with no corrupted records.

Analytics Pipeline: Data extraction from CSV → Pandas aggregation & data cleansing → Mean-difference telemetry signal modeling & Pearson correlation analysis → Executive decision support mapping.